

# Investor Protection and Firm Liquidity

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## ABSTRACT

The purpose of this study is to investigate the relation between investor protection and firm liquidity. We posit that less protective environments lead to wider bid-ask spreads and thinner depths because they fail to minimize information asymmetries. The Hong Kong equity market provides a unique opportunity to compare liquidity costs across distinct investor protection environments, but still within a common trading mechanism and currency. Our empirical findings verify that firm liquidity is significantly affected by investor protection. Regression and matched-sample results show that Hong Kong-based equities exhibit narrower spreads and thicker depths than their China-based counterparts.

THE PURPOSE of this study is to investigate the relation between economy-level investor protection and firm-level liquidity. We posit an inverse relation between the quality of investor protection and the cost of liquidity. A good investor protection environment minimizes information asymmetries and reduces the probability of trading against informed traders. In such an environment, liquidity providers incur relatively low costs and therefore offer narrow bid-ask spreads and thick depths. This is significant because lower liquidity costs have been shown to reduce the firm's cost of capital and increase its market value.<sup>1</sup> To date, the liquidity hypothesis has been difficult to test empirically because firms operating in distinct investor protection environments also trade on different exchanges, thus complicating the task of isolating investor protection effects from microstructure effects. We overcome this obstacle by holding constant the market structure and allowing investor protection levels to vary. The Hong Kong equity market lists firms from very distinct investor protection environments, but traded in the same market structure. Using the unique features of the Hong Kong market, we test and confirm the hypothesized relation between investor protection and firm liquidity.

Recent research demonstrates the importance of an economy's legal system on the quality of its investor protection. A market-oriented legal system reduces in-

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<sup>1</sup> See Amihud and Mendelson (1986), Barclay and Smith (1988), and Jacoby, Fowler, and Gottesman (2000) for theoretical expositions, and Chalmers and Kadlec (1998), Datar, Naik, and Radcliffe (1998), and Brennan and Subrahmanyam (1996) for supporting empirical evidence.

vestor uncertainty by establishing clear ownership rights, contract laws, commercial and bankruptcy codes, and by maintaining a high degree of certainty with respect to enforcement. The legal/regulatory environment largely determines the quantity and reliability of publicly available information, particularly at the firm level. By protecting investors from (unnecessary) information asymmetries, a well-developed legal framework enhances financial market liquidity. This view is consistent with Akerlof's (1970, p. 497) observation that "markets in underdeveloped countries often strongly reflect the operation of the Lemons Principle." Our study investigates the "operation of the Lemons Principle" in the liquidity provision process.

Previous studies show that secure investor rights encourage the development and growth of financial markets (La Porta et al. (2000)), and sophisticated financial markets produce higher rates of economic growth (King and Levine (1993), Levine (1997), Rajan and Zingales (1998)). Although this linkage from the legal system to investor protection, financial market development, and economic growth is intuitively appealing, considerable empirical work is necessary to pinpoint the mechanisms through which this chain of causation operates. At the economy-wide (macroliquidity) level, La Porta et al. (1997) and Levine (1999) confirm the positive relation between investor protection and various measures of capital market development. La Porta et al. (2002) find a positive relation between investor protection and Tobin's  $Q$ , and Wurgler (2000) shows that well-developed financial markets allocate capital more efficiently than less-developed markets. In this study, we investigate the relation between the quality of investor protection and firm (microlevel) liquidity. This approach, in contrast to previous studies, allows us to focus on the link between investor protection and asymmetric information. We expect that relatively strong investor protection will act to minimize the costs of asymmetric information, including the costs of liquidity. The empirical implication is that firms operating in countries with strong investor protection, *ceteris paribus*, will exhibit lower bid-ask spreads and thicker depths than firms operating in poor protection environments.<sup>2</sup>

The impact of investor protection on firm liquidity is an open empirical issue, although recent evidence is suggestive. Bhattacharya and Daouk (2002) show a reduction in the cost of equity following the enforcement (as opposed to the enactment) of insider trading rules across a broad cross section of countries. Lombardo and Pagano (2001) find some evidence of an inverse relation between shareholder rights and required rates of return on monthly equity indices. An inverse relation between investor protection and equity returns is consistent with an inverse relation between investor protection and liquidity costs. More directly, Jain (2001) investigates the relation between institutional design and firm liquidity. Although his study focuses primarily on variables under the control of the exchange (i.e., the market making system), one relevant result is that closing bid-ask spreads are narrower in countries with better shareholder rights.

<sup>2</sup>The positive relation between asymmetric information costs and bid-ask spreads is well documented in the literature. See Brockman and Chung (1999) and Heflin and Shaw (2001) for evidence of the inverse relation between asymmetric information costs and depths.

In this study, we analyze spreads and depths using 16 months of high-frequency data from the Stock Exchange of Hong Kong (SEHK).<sup>3</sup> The SEHK provides an appropriate setting for testing the liquidity hypothesis because it lists companies from very different investor protection environments. Under British colonial rule, the SEHK listed blue chip stocks as exemplified by the 33 constituents of the Hang Seng Index (HSI). Most blue chips originate and operate in an investor protection environment comparable to that of Western Europe or North America. More recently, the combination of China's ongoing commitment to "market socialism" and Hong Kong's status as a Special Administrative Region (SAR) of China has led to a large number of mainland Chinese listings, known as red chips and H-shares. While Hong Kong's investor protection environment is often ranked among the best in the world, China's investor protection is usually ranked at the opposite end of the spectrum.<sup>4</sup> Hong Kong's higher per capita wealth contributes to investor protection by improving the quality of legal and regulatory enforcement. We utilize this contrast between the Hong Kong and Chinese environments to rank firms according to their level of investor protection.

The distinct origins and operating environments of the blue chip and China-based firms lead to a straightforward ranking based on investor protection quality. Blue chips have a high level of investor protection because their assets, liabilities, revenues, expenses, and management teams have the least exposure to the Chinese investor protection environment. For the same reasons, China-based red chips and H-shares have a low level of investor protection. The ability to place firms into two distinct investor protection categories provides a unique framework in which to test our liquidity hypothesis. The empirical implication of this hypothesis is that firms subject to higher (lower) quality investor protection will have lower (higher) liquidity costs.

Our empirical findings confirm the posited relation between investor protection and firm liquidity. Regression analysis shows that lower quality investor protection induces higher liquidity costs (i.e., wider bid-ask spreads and thinner depths). The regression results are unaffected by the inclusion of various control variables including price, volume, and volatility. We also construct matched samples based on market capitalization and dollar volume to control for potential differences related to size and trading activity. The results from both matched samples are consistently supportive of the liquidity hypothesis.

Our study contributes to the financial literature by providing additional empirical evidence on the economic importance of investor protection. Our approach is comparable to recent studies that analyze investor protection issues

<sup>3</sup>In 2000, the SEHK, Hong Kong Futures Exchange Limited, and Hong Kong Securities Clearing Company Limited merged to form a single holding company, the Hong Kong Exchanges and Clearing Limited (HKEx). We will continue to refer to the Exchange as the SEHK because the merger occurred subsequent to our period of investigation.

<sup>4</sup>According to *The Economic Freedom of the World: 1998/1999 Interim Report* (1998), Hong Kong has the freest economy in the world, followed by Singapore, New Zealand, the United States, and the United Kingdom. China, on the other hand, ranks 79th in the world, just above Morocco, Fiji, and Papua New Guinea. Similarly, Hong Kong ranks 1st out of 156 countries in the 1998 *Index of Economic Freedom* report, while China ranks 107th.

using relatively detailed data from one or two countries (e.g., Zingales (1994) and Johnson (1999)), as opposed to coarser data from a large number of countries (e.g., La Porta et al. (1997)). Glaeser, Johnson, and Shleifer (2001), for example, conduct a “natural experiment” between two ex-socialist countries under very different regulatory regimes. They show that financial markets developed more rapidly under the stricter Polish regulatory environment relative to the less stringent Czech Republic approach. Similarly, our study investigates two economies with very different levels of investor protection. We show that diminished firm liquidity is one of the economic costs of poor investor protection. At the firm level, higher liquidity costs raise the firm’s cost of capital and reduce its market value. Across the economy, higher capital costs and lower valuations discourage investment and capital market development.

The remainder of the paper is organized as follows. Section I presents background information on the SEHK market structure and describes the spread and depth data. Section II discusses the method of analysis and analyzes the empirical results. Section III summarizes and concludes the paper.

## I. SEHK Market Structure and Data

### A. Market Structure

The order-driven market making system of the SEHK is very simple. There are no liquidity providers of last resort, no affirmative obligations to supply bid-ask quotations, no circuit breakers or other trading halts, no maximum price changes, and no exchange-designated order processors (e.g., *saitori*). Unlike the NYSE, AMEX, London Stock Exchange, and Tokyo Stock Exchange, the SEHK does not open with a call market and then switch to a continuous market; rather, it opens as a continuous market and remains a continuous market up to and including the close of trading.<sup>5</sup> All orders are placed through the public limit order book, and this information is instantaneously displayed on the trading terminals of various market participants. The only type of order permitted on the exchange is a limit order, so there are no complicated priority rules for market order execution as found on other automated exchanges such as the Paris Bourse (Biais, Hillion, and Spatt (1995)). In this sense, the SEHK is a very open and competitive market, particularly with respect to the ease of entry and exit for liquidity suppliers.

Order entry and execution begins with the submission of a limit order. The order is entered into the Automatic Order Matching and Execution System (AMS), which prioritizes it by price and then by time. Although order sizes are posted for each bid or ask price level, trade size is not a priority in execution. Bid orders are arranged in priority from highest to lowest and ask orders are arranged from lowest to highest. Since it is not possible to place a buy (sell) order above (below) the currently prevailing lowest ask (highest bid) price, the “best” price is the only one that can be executed at any given time. The quoted (and effective) bid-ask

<sup>5</sup> Our description of the SEHK reflects the rules in effect during our period of investigation.

spread is simply the difference between the prevailing lowest ask and highest bid price. Exchange members who trade on the floor of the Exchange, as well as non-members who have direct (real-time) access to the same information through data vendors, are free to post higher bids or lower offers as they see fit.

### *B. Data*

Our data set focuses on two distinct groups of stocks: blue chips and China-related firms (including red chips and H-shares). Blue chips, such as Henderson Land Development Co. Ltd., generally originate and operate in a favorable investor protection environment. H-shares, such as Tsingtao Brewery, are shares of mainland companies incorporated in China under Chinese law and later granted permission to raise capital in Hong Kong. These companies are former state-run monopolies with assets, liabilities, revenues, and expenses firmly entrenched in China. Although H-shares trade under the SEHK's rules and regulations, they operate in the Chinese investor protection environment. Red chips, such as CITIC Pacific, refer to holding companies usually set up by Chinese ministries or provincial governments. These firms often go public on the SEHK through a "back-door listing," whereby the mainland enterprise acquires a small, listed Hong Kong company and then injects it with a combination of mainland- and Hong Kong-acquired assets.

The data set is obtained from the SEHK's Research and Planning Division and includes intraday data for 33 blue chip stocks, 32 red chip stocks, and 34 H-shares covering the period from May 1, 1996, to August 29, 1997 (a total of 16 months).<sup>6</sup> The blue chip, red chip, and H-share categories correspond to the Hang Seng Index, Hang Seng China-Affiliated Corporations Index, and Hang Seng China-Enterprises Index, respectively, as of August 29, 1997. After dropping two firms from the sample because they are included in both the blue chip and red chip indexes, the final sample sizes are 31 blue chips and 64 China-based firms (30 red chips and 34 H-shares).<sup>7</sup> Our basic input data consist of stock prices, bid-ask spreads, and depths compiled at five-minute intervals throughout the trading day.

## **II. Empirical Results**

Table I presents summary market statistics for the two portfolios. The sample covers 330 trading days and 95 individual companies. The blue chips have a larger average market capitalization of HK\$70,780,190,885 compared to the China-based shares' HK\$3,795,131,815. These size differences suggest the need for subsequent

<sup>6</sup> Minor adjustments are made to the time-of-day for the first eight months of the sample period due to an internal clock misalignment in the original data capturing process. These adjustments are made based on information provided by SEHK's Research and Planning officials and verified by our program filters.

<sup>7</sup> The two companies are Guangdong Investment Ltd. and China Resources Enterprise Ltd. We replicate all empirical tests after including these firms in the blue chip category only and then again after including them in the China-based category only. Neither combination yields conclusions different from those reported herein.

**Table I**  
**Selected Summary Market Statistics on Trading Activities of the**  
**Sample Companies**

Two samples consisting of 31 blue chip Hong Kong-based companies and 64 China-based companies with ordinary shares listed on the Stock Exchange of Hong Kong (SEHK) over a 330-day sample period from May 1, 1996, to August 29, 1997, are included in this study. All measures of monthly averages are first calculated at the individual firm level and then averaged across all firms in each sample. Two trading days, October 14, 1996, and December 12, 1996, are excluded from the sample period because data on the bid and ask quotes were not available from the SEHK for these two particular days.

|                                                    | Blue Chip Sample   | China-based Sample |
|----------------------------------------------------|--------------------|--------------------|
| Number of companies in the sample                  | 31                 | 64                 |
| Average market capitalization                      | HK\$70,780,190,885 | HK\$3,795,131,815  |
| Average monthly trading volume in number of shares | 86,956,895         | 303,237,078        |
| Average monthly trading volume in dollar volume    | HK\$2,587,479,025  | HK\$876,227,849    |
| Average share price                                | HK\$34.3427        | HK\$3.1250         |
| Average monthly return                             | 0.0134             | 0.0843             |
| Average monthly volatility                         | 0.0073             | 0.0381             |
| Average absolute bid-ask spread                    | HK\$0.1254         | HK\$0.0280         |
| Average relative bid-ask spread                    | 0.0044             | 0.0126             |
| Average volume depth                               | 369,753            | 801,716            |
| Average dollar depth                               | HK\$10,530,043     | HK\$1,972,907      |

testing based on matched samples. Each of the two portfolios is actively traded with average monthly share volumes of 86,956,895 and 303,237,078, and average monthly dollar volumes of HK\$2,587,479,025 and HK\$876,227,849, for the blue chips and China-based shares, respectively. The higher share volumes yet lower dollar volumes for the Chinese firms are due to their lower average prices (HK\$3.1250) compared to blue chips (HK\$34.3427). The China-based firms have higher average monthly returns (0.0843 versus 0.0134) and volatility (0.0381 versus 0.0073) compared to the blue chips.

Table I also exhibits average liquidity values including two depth measures as well as absolute and relative bid-ask spreads. The average absolute (relative) spreads are HK\$0.1254 (0.0044) for blue chips, and HK\$0.0280 (0.0126) for the China-based shares. The difference in average prices explains the change in the relative rankings of absolute bid-ask spreads, defined as ask price minus bid price, and relative bid-ask spreads (*RBA*), defined as absolute spread divided by the bid-ask midpoint. Because relative spreads are more relevant from the investor's perspective, subsequent results are stated in terms of this measure. Volume depth is defined as the number of shares at the highest bid plus the number of shares at the lowest offer (i.e., the greatest number of shares that can be bought or sold without moving prices). Dollar depth (*DDepth*) is the number of shares at the bid and ask multiplied by their respective prices. The dollar depths (volume depths) are HK\$10,530,043 (369,753) for blue chips, and HK\$1,972,907 (801,716) for the China shares.

Figure 1 presents the relation between relative spreads and monthly return volatility. The blue chip firms tend to cluster in the lower, left-hand side of the

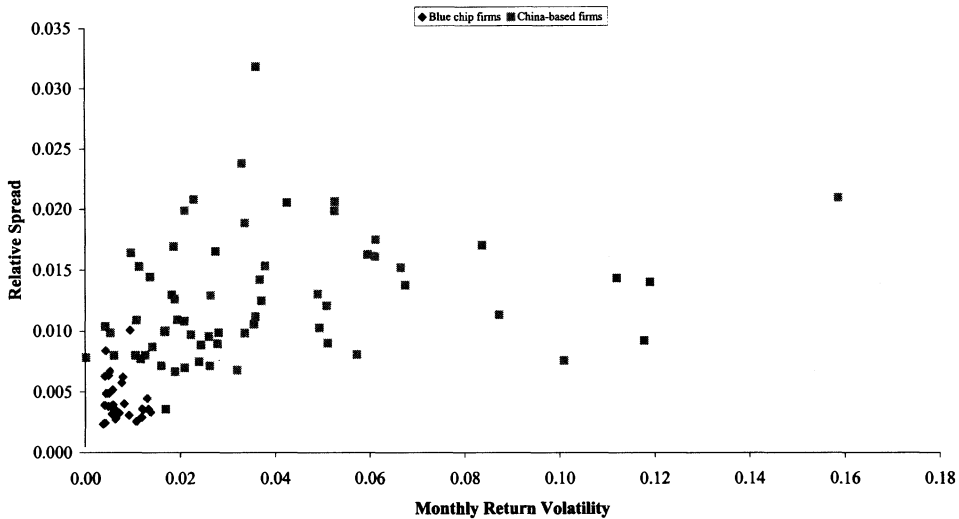


Figure 1. Relative spread and monthly return volatility.

graph. They display not only lower spreads and volatility than their China-based counterparts, but also lower spreads for an equivalent amount of volatility. Figure 2 presents the relation between relative spreads and monthly trading volume. Again, the blue chip firms tend to cluster in the lower, left-hand side of the graph, signifying lower spreads for an equivalent amount of volume. The spread/volatility and spread/volume relations are consistent with expectations based on the liquidity hypothesis (i.e., weaker investor protection imposes higher liquidity costs for a given level of volatility or volume).

#### A. Regression Results

Following McNish and Wood (1992) and others, we calculate our two liquidity measures, *RBA* and *DDepth*, using time-weighted averages. The monthly liquidity measures are time weighted at five-minute intervals over each calendar month.<sup>8</sup> We estimate a pooled cross-sectional, time-series regression using all available firm-months. Two separate regressions, one for each of our two liquidity measures, are estimated as follows:

$$Liquidity_{i,t} = \sum_{j=1}^{31} \alpha_j bluechip_{i,t}^j + \sum_{j=32}^{95} \beta_j china_{i,t}^j + \varepsilon_{i,t} \quad (1)$$

where Liquidity refers to either *RBA* or *DDepth* (in logarithms) as defined above. In this specification, each firm's mean liquidity value is estimated separately. We

<sup>8</sup> We also specify *RBA* and *DDepth* based on (1) all five-minute intervals (i.e., no averaging), (2) five-minute intervals time-weighted over daily periods, (3) five-minute intervals time-weighted over weekly periods, and (4) firm-level averages. The regression results reported herein are robust to all such alternative specifications.

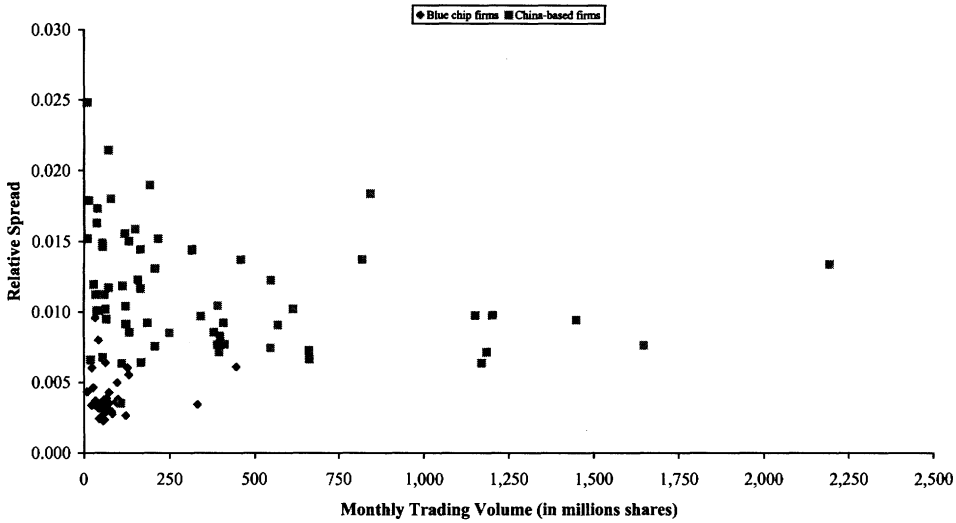


Figure 2. Relative spread and monthly trading volume.

then compare the blue chip group mean to the China-based group mean under the null hypothesis of equality (i.e.,  $\frac{1}{31} \sum_{j=1}^{31} \hat{\alpha}_j = \frac{1}{64} \sum_{j=32}^{95} \hat{\beta}_j$ ). An alternative specification using group dummy variables would restrict all firms within a group to have the same mean liquidity measure. Since both specifications yield the same overall conclusion, we report only the findings from the less restrictive model. The purpose of this and all subsequent regressions is to capture liquidity differences caused by different protection levels. Consistent with the liquidity hypothesis, we expect the mean of the blue chip coefficients for *RBA* to be significantly lower than the corresponding mean of the China-based coefficients. We also expect the mean of the blue chip coefficients for *DDepth* to be significantly higher than the corresponding China-based means.

Table II presents the results for the first set of regressions without control variables. The mean *RBA* coefficient for the blue chip group is  $-5.5533$  (in logarithms) and the corresponding mean *RBA* coefficient for the China-based group is  $-4.5575$ . More importantly, the test for equality of mean coefficients rejects the null hypothesis at the one percent level. The mean *DDepth* coefficient for blue chips is  $15.5293$  and the corresponding coefficient for the China-based firms is  $13.7427$ . As before, the *F*-statistic is significant at the one percent level. The *RBA* and *DDepth* measures of liquidity show that blue chips are more liquid than the China-based firms. Because these two dimensions of liquidity point in the same direction, we can unambiguously rank blue chips and China-based firms in terms of liquidity (Lee, Mucklow, and Ready (1993)). These preliminary findings are consistent with the hypothesis that differences in investor protection lead to differences in liquidity.

Previous research demonstrates that bid-ask spreads are at least partly determined by variations in volume, volatility, and price (e.g., see Benston and Hagerman (1974), Stoll (1978), and Barclay and Smith (1988)). It is therefore plausible

Table II  
Differences in Firm Liquidity Across Distinct Investor Protection Environments

Regression of liquidity measures on the Hong Kong blue chip and the China-based variables and testing for differences between the blue chip group mean and the China-based group mean:

$$Liquidity_{i,t} = \sum_{j=1}^{31} \alpha_j bluechip^j_{i,t} + \sum_{j=32}^{95} \beta_j china^j_{i,t} + \varepsilon_{i,t}$$

where  $Liquidity_{i,t}$  is a liquidity measure of firm  $i$  for month  $t$ , representing  $RBA_{i,t}$  or  $DDepth_{i,t}$ . The first dependent variable,  $RBA_{i,t}$ , is the relative bid-ask spread for a firm, measured as the difference between the lowest ask and the highest bid divided by the bid-ask midpoint. The second dependent variable,  $DDepth_{i,t}$ , is the dollar depth for a firm, which is the dollar value of all shares quoted at the lowest ask plus the dollar value of all shares quoted at the highest bid. The liquidity variables are all time-weighted across the intraday intervals to generate an overall monthly average for each variable. The variables,  $Bluechip^j$ ,  $j = 1, \dots, 31$ , represent a set of dummy variables for the 31 blue chip companies and each variable is assigned a value of one if firm  $i$  is the Hong Kong blue chip company  $j$ , and zero otherwise; the subscripts are dropped for notational convenience. Similarly,  $china^j$ ,  $j = 32, \dots, 95$ , represent a set of dummy variables for the 64 China-based companies and each variable is assigned a value of one if firm  $i$  is the China-based company  $j$ , and zero otherwise. All non-dummy variables,  $RBA_{i,t}$  and  $DDepth_{i,t}$ , are transformed by taking natural logarithms. Total number of observations for the regression is 1,362 representing all firm-months with data available for the sample companies over the sample period. This table shows the average estimated coefficients and the average  $t$ -statistics for the 31 *bluechip*<sup>*j*</sup> variables and the 64 *china*<sup>*j*</sup> variables; estimated coefficients and  $t$ -statistics of the individual dummy variables (i.e., *bluechip*<sup>*j*</sup>,  $j = 1, \dots, 31$ , and *china*<sup>*j*</sup>,  $j = 32, \dots, 95$ ) are not reported.

| Variable                                                 | Liquidity <sub>it</sub> = RBA <sub>it</sub>   |                     | Liquidity <sub>it</sub> = DDepth <sub>it</sub> |                     |
|----------------------------------------------------------|-----------------------------------------------|---------------------|------------------------------------------------|---------------------|
|                                                          | Estimated Coefficient                         | <i>t</i> -statistic | Estimated Coefficient                          | <i>t</i> -statistic |
| <i>Bluechip</i> <sup><i>j</i></sup> , $j = 1, \dots, 31$ | $\frac{1}{31} \sum_{j=1}^{31} \hat{\alpha}_j$ | - 5.5533            | - 90.07 <sup>a</sup>                           | 103.53 <sup>a</sup> |
| <i>China</i> <sup><i>j</i></sup> , $j = 32, \dots, 95$   | $\frac{1}{64} \sum_{j=32}^{95} \hat{\beta}_j$ | - 4.5575            | - 65.98 <sup>a</sup>                           | 81.71 <sup>a</sup>  |

Tests for equality of the average coefficients:

$$H_0 : \frac{1}{31} \sum_{j=1}^{31} \hat{\alpha}_j = \frac{1}{64} \sum_{j=32}^{95} \hat{\beta}_j$$

*F*-statistic (d.f. = 1 and 1,267)

\*Test statistic significant at the 0.01 level in a one- or two-tail test (as appropriate).

<sup>a</sup>An average of the  $t$ -statistics of all the estimated coefficients in the set; all of the  $t$ -statistics are significant at the 0.01 level.

that the significant liquidity differences reported in Table II are due to variations in these three explanatory variables and not to the different investor protection environments. To explore this possibility, the following fixed-effects regression model is estimated with volume, volatility, and price as control variables:

$$Liquidity_{i,t} = \sum_{j=1}^{31} \alpha_j bluechip_{i,t}^j + \sum_{j=32}^{95} \beta_j china_{i,t}^j + \gamma_1 vol_{i,t} + \gamma_2 var_{i,t} + \gamma_3 price_{i,t} + \varepsilon_{i,t} \quad (2)$$

where  $vol_{i,t}$  is defined as the total number of shares traded for firm  $i$  over month  $t$ ,  $price_{i,t}$  is the average stock price for firm  $i$  over month  $t$ , and  $var_{i,t}$  is the stock return volatility for firm  $i$  over month  $t$ . All variables,  $RBA$ ,  $DDepth$ ,  $vol$ ,  $var$ , and  $price$ , are transformed by taking their natural logarithms. The fixed-effects model is robust to omitted time-invariant regressors and, therefore, recommended when individual effects ( $\alpha_j$ 's and  $\beta_j$ 's) are likely to be correlated with the explanatory variables (see Johnston and DiNardo (1997)).

Table III presents the results from estimating equation (2). The signs of the control variables are consistent with theory and previous empirical findings. Higher trading volumes are expected to reduce relative bid-ask spreads ( $\gamma_1 = -0.0965$ ) and increase dollar depths ( $\gamma_1 = 0.4832$ ) as higher volume levels allow market makers to spread their fixed costs over more units. Higher return variance is expected to increase spreads ( $\gamma_2 = 0.0286$ ) and reduce dollar depths ( $\gamma_2 = -0.1232$ ) as greater volatility increases the costs of inventory control. Higher prices are associated with lower relative spreads ( $\gamma_3 = -0.3607$ ) because of a fixed cost component of the spread (i.e., there is less variation in market making costs than prices). Higher prices are associated with higher  $DDepths$  ( $\gamma_3 = 0.5625$ ), suggesting that one can transact larger dollar amounts at prevailing bid and ask prices for higher priced stocks than for lower priced stocks. Overall, the estimated coefficients, in terms of signs, magnitudes, and levels of statistical significance, confirm theory-based expectations.

The most significant result in Table III, however, is that both regression models confirm the hypothesized liquidity ranking of blue chips and China-based firms. In terms of  $RBA$ , the blue chip average coefficient of  $-2.4975$  is significantly lower than the corresponding China-based coefficient of  $-2.3402$ . In terms of  $DDepth$ , the blue chip average coefficient of  $4.2468$  is significantly higher than the corresponding China-based coefficient of  $3.6917$ . Therefore, the inclusion of various control variables based on market microstructure theory does not alter the previous conclusion that the Hong Kong-based investor environment is associated with higher levels of liquidity.<sup>9</sup>

<sup>9</sup>In addition to using alternative time intervals (e.g., daily, weekly) and control variables (e.g., market capitalization, dollar volume), we also test alternative estimation methods including OLS with group dummy variables and Park's (1967) method to account for serially and contemporaneously correlated disturbances. All estimations are also repeated using volume depths as the dependent variable. The results are consistent with those reported above.

Table III  
Differences in Firm Liquidity Across Distinct Investor Protection Environments: Additional Control Variables  
Regression of liquidity measures on the Hong Kong blue chip and the China-based variables controlling for company differences and the effects of other control variables in a fixed-effect panel data model:

$$Liquidity_{i,t} = \sum_{j=1}^{31} \alpha_j bluechip^j_{i,t} + \sum_{j=32}^{95} \beta_j china^j_{i,t} + \gamma_1 vol_{i,t} + \gamma_2 var_{i,t} + \gamma_3 price_{i,t} + \varepsilon_{i,t},$$

where  $Liquidity_{i,t}$  is a liquidity measure of firm  $i$  for month  $t$ , defined in the same way as in Table II. The variables,  $Bluechip^j$ ,  $j = 1, \dots, 31$ , represent a set of dummy variables for the 31 blue chip companies and each variable is assigned a value of one if firm  $i$  is the Hong Kong blue chip company  $j$ , and zero otherwise; the subscripts are dropped for notational convenience. Similarly,  $china^j$ ,  $j = 32, \dots, 95$ , represent a set of dummy variables for the 64 China-based companies and each variable is assigned a value of one if firm  $i$  is the China-based company  $j$ , and zero otherwise. The control variables are  $vol_{i,t}$ ,  $var_{i,t}$ , and  $price_{i,t}$ , for volume, volatility, and price of firm  $i$  over month  $t$ , where  $vol_{i,t}$  is the total number of shares traded during the month,  $var_{i,t}$  is the volatility of returns calculated across all weeks over the month, and  $price_{i,t}$  is the average transaction price during the month. All non-dummy variables,  $vol_{i,t}$ ,  $var_{i,t}$ , and  $price_{i,t}$ , are transformed by taking natural logarithms. Total number of observations for the regression is 1,362, representing all firm-months with data available for the sample companies over the sample period. This table shows the average estimated coefficients and the average  $t$ -statistics for the 31 *bluechip*<sup>*j*</sup> variables and the 64 *china*<sup>*j*</sup> variables; estimated coefficients and  $t$ -statistics of the individual dummy variables (i.e.,  $bluechip^j$ ,  $j = 1, \dots, 31$ , and  $china^j$ ,  $j = 32, \dots, 95$ ) are not reported.

| Variable                          | $Liquidity_{i,t} = RBA_{i,t}$                 |                     |  | $Liquidity_{i,t} = DDepth_{i,t}$ |                     |  |
|-----------------------------------|-----------------------------------------------|---------------------|--|----------------------------------|---------------------|--|
|                                   | Estimated Coefficient                         | $t$ -statistic      |  | Estimated Coefficient            | $t$ -statistic      |  |
| $Bluechip^j$ , $j = 1, \dots, 31$ | $\frac{1}{31} \sum_{j=1}^{31} \hat{\alpha}_j$ | -16.62 <sup>a</sup> |  | 4.2468                           | 15.26 <sup>a</sup>  |  |
| $China^j$ , $j = 32, \dots, 95$   | $\frac{1}{64} \sum_{j=32}^{95} \hat{\beta}_j$ | -14.17 <sup>a</sup> |  | 3.6917                           | 12.04 <sup>a</sup>  |  |
| Trading volume                    | $\hat{\gamma}_1$                              | -12.06 <sup>*</sup> |  | 0.4832                           | 32.68 <sup>*</sup>  |  |
| Return variance                   | $\hat{\gamma}_2$                              | 6.08 <sup>*</sup>   |  | -0.1232                          | -14.17 <sup>*</sup> |  |
| Trading price                     | $\hat{\gamma}_3$                              | -19.74 <sup>*</sup> |  | 0.5625                           | 16.66 <sup>*</sup>  |  |

Tests for equality of the average coefficients:

$$H_0 : \frac{1}{31} \sum_{j=1}^{31} \hat{\alpha}_j = \frac{1}{64} \sum_{j=32}^{95} \hat{\beta}_j$$

F-statistic (d.f. = 1 and 1,264)

10.88<sup>\*</sup>

39.71<sup>\*</sup>

<sup>\*</sup>Test statistic significant at the 0.01 level in a one- or two-tail test (as appropriate).

<sup>a</sup>An average of the  $t$ -statistics of all the estimated coefficients in the set; all of the  $t$ -statistics are significant at the 0.01 level.

### B. Matched Samples

Next, we construct two matched samples from the existing portfolios in order to further isolate the investor protection effect from other potentially confounding effects. For example, firm-specific liquidity differences not captured by volume, volatility, and price control variables could be responsible for observed liquidity differences between Hong Kong- and China-based companies. The quantity and quality of firm-specific differences are expected to be related to dollar trading volume and market capitalization (i.e., activity and size). If Hong Kong-based firms exhibit lower spreads and higher depths than China-based companies of equivalent dollar trading volume and market capitalization, then this is additional evidence in favor of a distinct investor protection effect.

The matched samples are constructed as follows: For each blue chip, we identify a China-based share that is within 10 percent of the blue chip's dollar trading volume (if any).<sup>10</sup> The construction process results in a sample of 46 companies: 23 blue chips and 23 China-based companies. The benefit of the matched sample approach is that it can control for latent factors, but it also imposes a cost of reduced sample size. Average dollar volumes are HK\$2,738,879,216 and HK\$2,738,757,775 for the blue chip and China-based portfolios, respectively. The market capitalization-matched sample is constructed in the same manner, with market capitalization substituted for dollar volume. Because the blue chips are larger than the China-based firms on average, the selection process tends to match relatively low market-value blue chips (among blue chips) to relatively high market-value red chips and H-shares (among China-based firms). This process results in a market capitalization-matched sample of 18 companies: 9 Hong Kong-based blue chips and 9 China-based companies. Average market values are HK\$19,829,763,048 and HK\$19,481,556,036 for the Hong Kong and China-based portfolios, respectively.

This matching procedure also addresses a potential issue related to "visibility." Firms belonging to the blue chip, red chip, and H-share categories receive more press coverage and investor interest than firms not belonging to a Hang Seng-designated index. By restricting our matches to firms within these high-profile portfolios, we mitigate the possibility that observed liquidity effects are due to visibility differences. Ultimately, we attempt to hold constant all variables that may explain cross-sectional liquidity variation (i.e., volume, volatility, price, visibility, market capitalization, and dollar volume) other than differences due to investor protection quality.

Tables IV and V report the matched-sample results using fixed-effects regression model (2).<sup>11</sup> In both tables, estimated control variable coefficients are consistent with theory. The China-based companies have significantly lower levels of

<sup>10</sup> Dollar volume matches are based on August 1997 monthly volume figures and market capitalization matches are based on August 29, 1997, closing prices and outstanding shares, as reported in the SEHK's *Securities Journal* (August, 1997 trading record edition).

<sup>11</sup> Again, we also test alternative time intervals (e.g., daily, weekly), control variables (e.g., market capitalization, dollar volume), and estimation methods (e.g., OLS with group dummy variables, Park's (1967) method). The results are consistent with those reported above.

Table IV  
Differences in Firm Liquidity across Distinct Investor Protection Environments: Dollar Trading  
Volume-matched Sample

Regression of liquidity measures on a sample of Hong Kong blue chip and China-based companies matched on the basis of dollar trading volume, controlling for company differences and the effects of other control variables in a fixed-effect panel data model:

Liquidity<sub>i,t</sub> = ∑<sub>j=1</sub><sup>23</sup> α<sub>j</sub> bluechip<sup>j</sup><sub>i,t</sub> + ∑<sub>j=24</sub><sup>46</sup> β<sub>j</sub> china<sup>j</sup><sub>i,t</sub> + γ<sub>1</sub> vol<sub>i,t</sub> + γ<sub>2</sub> var<sub>i,t</sub> + γ<sub>3</sub> price<sub>i,t</sub> + e<sub>i,t</sub>,

where *Liquidity<sub>i,t</sub>* is a liquidity measure of firm *i* for month *t*, defined in the same way as in Table II. The variables, *Bluechip<sup>j</sup>*, *j* = 1, ..., 23, represent a set of dummy variables for the 23 blue chip companies and each variable is assigned a value of one if firm *i* is the Hong Kong blue chip company *j*, and zero otherwise; the subscripts are dropped for notational convenience. Similarly, *china<sup>j</sup>*, *j* = 24, ..., 46, represent a set of dummy variables for the 23 China-based companies, and each variable is assigned a value of one if firm *i* is the China-based company *j*, and zero otherwise. The control variables are *vol<sub>i,t</sub>*, *var<sub>i,t</sub>*, and *price<sub>i,t</sub>* for volume, volatility, and price of firm *i* over month *t*, where *vol<sub>i,t</sub>* is the total number of shares traded during the month, *var<sub>i,t</sub>* is the volatility of returns calculated across all weeks over the month, and *price<sub>i,t</sub>* is the average transaction price during the month. All non-dummy variables, *vol<sub>i,t</sub>*, *var<sub>i,t</sub>*, and *price<sub>i,t</sub>*, are transformed by taking natural logarithms. Total number of observations for the regression is 674 representing all firm-months with data available for the sample companies over the sample period. This table shows the average estimated coefficients and the average *t*-statistics for the 23 *bluechip<sup>j</sup>* variables and the 23 *china<sup>j</sup>* variables; estimated coefficients and *t*-statistics of the individual dummy variables (i.e., *bluechip<sup>j</sup>*, *j* = 1, ..., 23, and *china<sup>j</sup>*, *j* = 24, ..., 46) are not reported.

| Variable                                            | Liquidity <sub>i,t</sub> = RBA <sub>i,t</sub> |                     | Liquidity <sub>i,t</sub> = DDepth <sub>i,t</sub> |                     |
|-----------------------------------------------------|-----------------------------------------------|---------------------|--------------------------------------------------|---------------------|
|                                                     | Estimated Coefficient                         | <i>t</i> -statistic | Estimated Coefficient                            | <i>t</i> -statistic |
| <i>Bluechip<sup>j</sup></i> , <i>j</i> = 1, ..., 23 | $\frac{1}{23} \sum_{j=1}^{23} \hat{\alpha}_j$ | -3.1551             | -15.09 <sup>a</sup>                              | 4.2844              |
| <i>China<sup>j</sup></i> , <i>j</i> = 24, ..., 46   | $\frac{1}{23} \sum_{j=24}^{46} \hat{\beta}_j$ | -2.8681             | -13.03 <sup>a</sup>                              | 3.5165              |
| Trading volume                                      | $\hat{\gamma}_1$                              | -0.0723             | -6.47 <sup>*</sup>                               | 0.5054              |
| Return variance                                     | $\hat{\gamma}_2$                              | 0.0211              | 3.43 <sup>*</sup>                                | -0.1223             |
| Trading price                                       | $\hat{\gamma}_3$                              | -0.3241             | -11.17 <sup>*</sup>                              | 0.3792              |

Tests for equality of the average coefficients:

H<sub>0</sub> :  $\frac{1}{23} \sum_{j=1}^{23} \hat{\alpha}_j = \frac{1}{23} \sum_{j=24}^{46} \hat{\beta}_j$

F-statistic (d.f. = 1 and 625)

19.71<sup>\*</sup>

38.87<sup>\*</sup>

<sup>\*</sup>Test statistic significant at the 0.01 level in a one- or two-tail test (as appropriate).

<sup>a</sup>An average of the *t*-statistics of all the estimated coefficients in the set; all of the *t*-statistics are significant at the 0.01 level.

Table V  
Differences in Firm Liquidity across Distinct Investor Protection Environments: Market Capitalization-matched Sample

Regression of liquidity measures on a sample of Hong Kong blue chip and China-based companies matched on the basis of market capitalization, controlling for company differences and the effects of other control variables in a fixed-effect panel data model:

$$Liquidity_{it} = \sum_{j=1}^9 \alpha_j bluechip^j_{it} + \sum_{j=10}^{18} \beta_j china^j_{it} + \gamma_1 vol_{it} + \gamma_2 vari_{it} + \gamma_3 price_{it} + \varepsilon_{it},$$

where  $Liquidity_{it}$  is a liquidity measure of firm  $i$  for month  $t$ , defined in the same way as in Table II. The variables,  $Bluechip^j$ ,  $j = 1, \dots, 9$ , represent a set of dummy variables for the nine blue chip companies, and each variable is assigned a value of one if firm  $i$  is the Hong Kong blue chip company  $j$ , and zero otherwise; the subscripts are dropped for notational convenience. Similarly,  $china^j$ ,  $j = 10, \dots, 18$ , represent a set of dummy variables for the nine China-based companies, and each variable is assigned a value of one if firm  $i$  is the China-based company  $j$ , and zero otherwise. The control variables are  $vol_{it}$ ,  $vari_{it}$ , and  $price_{it}$ , for volume, volatility, and price of firm  $i$  over month  $t$ , where  $vol_{it}$  is the total number of shares traded during the month,  $vari_{it}$  is the volatility of returns calculated across all weeks over the month, and  $price_{it}$  is the average transaction price during the month. All non-dummy variables,  $vol_{it}$ ,  $vari_{it}$ , and  $price_{it}$ , are transformed by taking natural logarithms. Total number of observations for the regression is 287, representing all firm-months with data available for the sample companies over the sample period. This table shows the average estimated coefficients and the average  $t$ -statistics for the nine *bluechip* variables and the nine *china* variables; estimated coefficients and  $t$ -statistics of the individual dummy variables (i.e.,  $bluechip^j$ ,  $j = 1, \dots, 9$ , and  $china^j$ ,  $j = 10, \dots, 18$ ) are not reported.

| Variable                                                                                    | Liquidity <sub>it</sub> = RBA <sub>it</sub>  |                | Liquidity <sub>it</sub> = DDepth <sub>it</sub> |                    |
|---------------------------------------------------------------------------------------------|----------------------------------------------|----------------|------------------------------------------------|--------------------|
|                                                                                             | Estimated Coefficient                        | $t$ -statistic | Estimated Coefficient                          | $t$ -statistic     |
| $Bluechip^j$ , $j = 1, \dots, 9$                                                            | $\frac{1}{9} \sum_{j=1}^9 \hat{\alpha}_j$    | -3.8867        | 2.5896                                         | 3.72 <sup>a</sup>  |
| $China^j$ , $j = 10, \dots, 18$                                                             | $\frac{1}{9} \sum_{j=10}^{18} \hat{\beta}_j$ | -3.8095        | 2.2581                                         | 3.03 <sup>b</sup>  |
| Trading volume                                                                              | $\hat{\gamma}_1$                             | -0.0247        | 0.5893                                         | 15.03 <sup>*</sup> |
| Return variance                                                                             | $\hat{\gamma}_2$                             | 0.0053         | -0.1121                                        | -5.45 <sup>*</sup> |
| Trading price                                                                               | $\hat{\gamma}_3$                             | -0.4291        | 0.3971                                         | 5.97 <sup>*</sup>  |
| Tests for equality of the average coefficients:                                             |                                              |                |                                                |                    |
| $H_0: \frac{1}{9} \sum_{j=1}^9 \hat{\alpha}_j = \frac{1}{9} \sum_{j=10}^{18} \hat{\beta}_j$ |                                              |                |                                                |                    |
| $F$ -statistic (d.f. = 1 and 266)                                                           |                                              |                |                                                |                    |
|                                                                                             |                                              |                | 1.89                                           | 9.34 <sup>*</sup>  |

<sup>\*</sup>Test statistic significant at the 0.01 level in a one- or two-tail test (as appropriate).  
<sup>a</sup>An average of the  $t$ -statistics of all the estimated coefficients in the set; all of the  $t$ -statistics are significant at the 0.01 level.  
<sup>b</sup>An average of the  $t$ -statistics of all the estimated coefficients in the set; all except one of the  $t$ -statistics are significant at the 0.01 level.

liquidity in both the spread and depth dimensions compared to the dollar volume-matched, Hong Kong-based blue chips in Table IV. The average blue chip *RBA* coefficient of  $-3.1551$  is significantly smaller than its China-based counterpart of  $-2.8681$  at the one percent level of significance. The average blue chip *DDepth* coefficient of  $4.2844$  is significantly larger than its China-based counterpart of  $3.5165$ , again at the one percent level of significance.

Similar results are displayed in Table V for the capitalization-matched sample, except that the average blue chip *RBA* coefficient of  $-3.8867$  is not significantly lower than the China-based average coefficient of  $-3.8095$  at conventional levels of significance ( $p$ -value = 0.17). However, even with only nine firms from each investor protection group, the average blue chip *DDepth* coefficient is significantly larger than its China-based counterparts at the one percent level of significance. Overall, these results provide strong evidence for the existence of a liquidity effect uniquely associated with the investor environment.

The matched-sample findings demonstrate that market participants' willingness to place limit orders at higher bid or lower ask prices (i.e., to narrow the spread) or to increase the size of their orders in terms of dollar volume (i.e., to increase the depth) is significantly influenced by the firm's investor protection environment.

### III. Summary and Conclusion

Recent research suggests a chain of causation from a country's level of investor protection to financial market development, and from financial market development to economic growth. Although this conceptual framework is straightforward and intuitively appealing, the underlying mechanisms that bind one link to the next remain largely undocumented. The purpose of our study is to establish one of the connections from investor protection to financial market development. We posit that strong investor protection reduces the liquidity costs associated with asymmetric information. Market-supportive laws, regulations, and enforcement give rise to an investor protection environment that is conducive to the supply of liquidity by minimizing information costs. Consistent with this analysis, previous research (e.g., Levine (1999) and La Porta et al. (1997)) finds a positive relation between investor protection and macroliquidity (i.e., country-level debt and equity market values). We investigate the relation between investor protection and microliquidity (i.e., firm-level bid-ask spreads and depths).

To date, there is little empirical evidence linking investor protection to firm-level liquidity because firms from distinct investor protection environments tend to trade in different countries and on different exchanges. The SEHK is an ideal setting in which to test the liquidity hypothesis because of the Exchange's role in channeling international capital to the emerging Chinese market. The existence of blue chips and China-related red chips and H-shares allows us to rank portfolios of companies according to their level of investor protection based on their exposure to China's legal system. The empirical implication is that liquidity costs

will be lowest for the Hong Kong-based blue chips and highest for the China-based firms.

We test whether bid-ask spreads (depths) are relatively lower (higher) for the Hong Kong-based firms. The empirical results provide strong support for the liquidity hypothesis with respect to both spreads and depths. Subsequent testing shows that the liquidity ranking of blue chips and China-related firms is robust to the addition of various control variables (i.e., volume, volatility, and price level), and that Hong Kong-based firms are significantly more liquid than China-based firms matched on the basis of dollar volume and market capitalization. We conclude that China's weak investor protection environment reduces overall firm liquidity. La Porta et al. (1998, p. 1152) state that, "The ultimate question, of course, is whether countries with poor investor protections—either laws or their enforcement—actually do suffer." The results of our study show that one type of "suffering" comes in the form of higher liquidity costs.

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